

Weather Impact on US Public Transportation

A project report on how weather affects transit systems in Boston, Chicago, and New York.

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Executive Summary



Key Finding

Snowy weather causes 114% more delays than clear conditions.



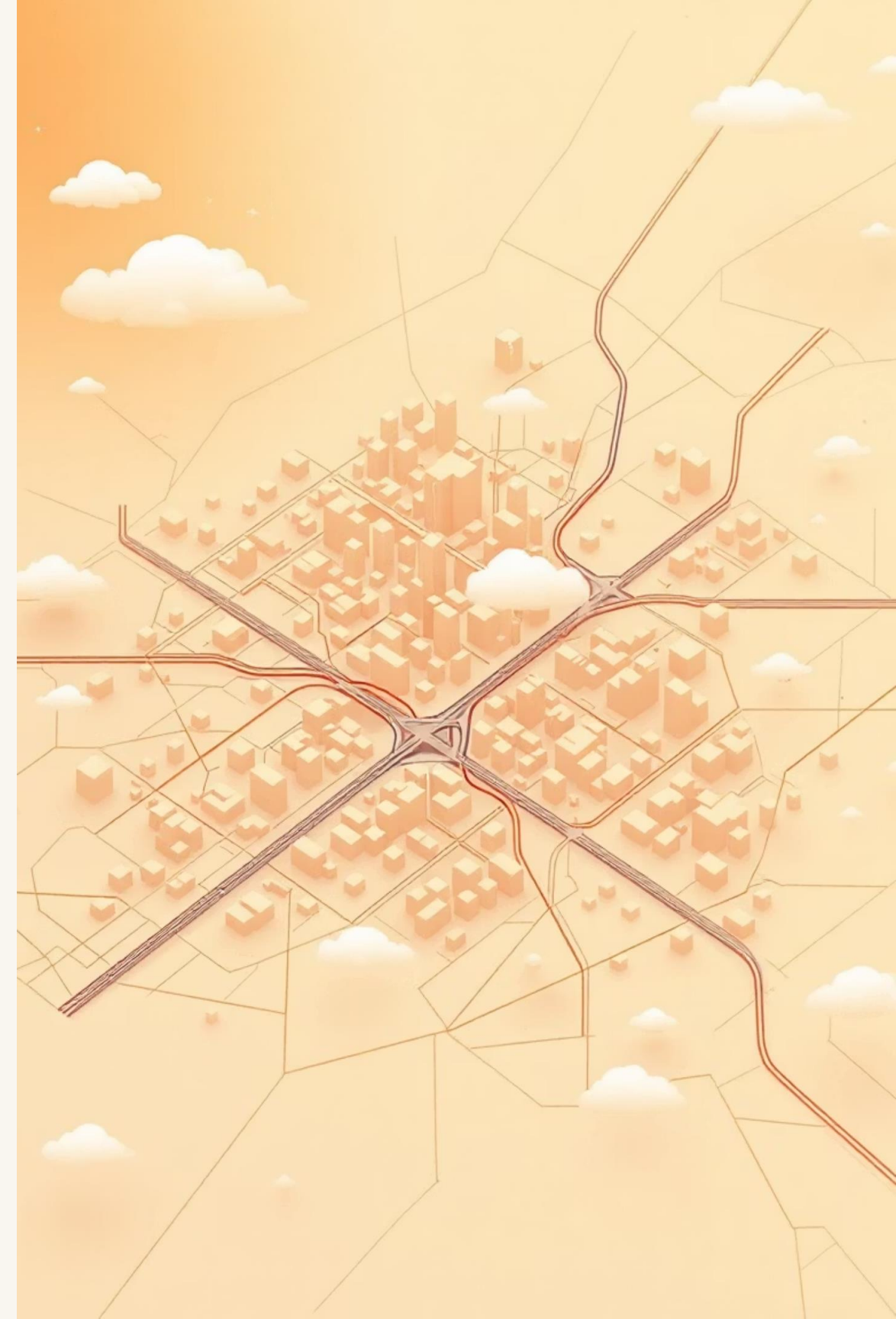
Coverage

Boston, Chicago, New York.



Data Analyzed

364,038 records across 429 routes over 731 days.



Project Overview

Problem Statement

US public transportation systems lose millions annually due to weather-related delays. Transit agencies lack quantitative data to:

- Predict disruptions
- Allocate resources efficiently
- Justify infrastructure investments
- Communicate expectations to passengers

Objectives

- Quantify weather impact
- Identify disruptive conditions
- Compare city resilience
- Build monitoring pipeline
- Deliver actionable insights



Data Collection & Processing

1

Weather Data

Visual Crossing Weather API:
Daily observations for **3 cities**
over **731 days**.

2

Transit Data

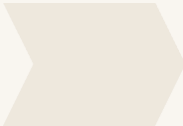
GTFS feeds: **143 routes** each for
MBTA, CTA, MTA (**429 total**).

3

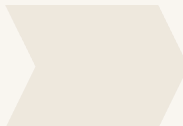
ETL Process

Extract, Transform, Load into PostgreSQL, including cleaning and merging data.

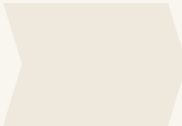
Pipeline Execution Flow



Airflow Scheduler
(Every 5 minutes)

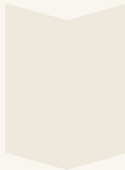


Trigger US Cities DAG



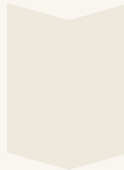
Step 1: Fetch Weather Data

- API Call to Visual Crossing
- Fetch data for 3 cities
- Save to weather_data.csv



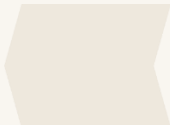
Step 2: Fetch Transit Data

- Fetching Transit Data (GTFS)
- Correlate with weather patterns
- Save to transit_data.csv



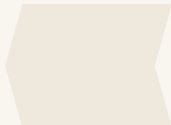
Step 3: Process Data

- Load CSV files
- Merge datasets
- Clean and validate
- Insert into PostgreSQL



Step 4: Run Analysis

- Query weather_db
- Calculate correlations
- Generate insights
- Log results



DAG Complete

Analysis Results: Overall Impact

Weather Condition	Avg Delay	% of Total	Impact vs Clear
Snowy	2.14 min	15.6%	+114.0%
Rainy	1.40 min	30.0%	+40.0%
Windy	1.06 min	10.8%	+6.0%
Clear	1.00 min	43.6%	Baseline

Snow causes more than double the delays, while rain increases delays by 40%.

San e 2024



Season

S	M	T	W	T	F	S
0	2	3	5	4	6	
6	7	8	16	15	10	
13	11	18	16	19	29	
15	15	27	25	23	27	
24	25	24				

Season

S	M	T	W	T	F	S
3	5	8	14	1	5	0
6	11	17	11	18	17	13
17	15	19	24	11	18	29
18	24	23	24	26	27	
23						

Monday

S	M	T	W	T	F	S
			1	8	2	4
6	7	8	0	17	11	15
16	17	13	19	15	13	17
18	28	12	27	25	21	27
23	21	23	28			



Tuesday

S	M	T	W	T	F	S
5	6	13	14	11	16	
6	15	14	13	15	16	
14	15	19	24	23	17	
23	25	26	29	25	28	

Monday

S	M	T	W	T	F	S
5	4	5	16	14	15	10
15	14	16	10	11	15	18
17	23	15	25	24	29	28
25	23	28				

Friday

S	M	T	W	T	F	S
5	4	5	6	10	11	15
10	16	15	18	15	15	29
23	23	23	18	29	29	27
23	27	23	24	29		

City-Specific & Seasonal Patterns

City Analysis

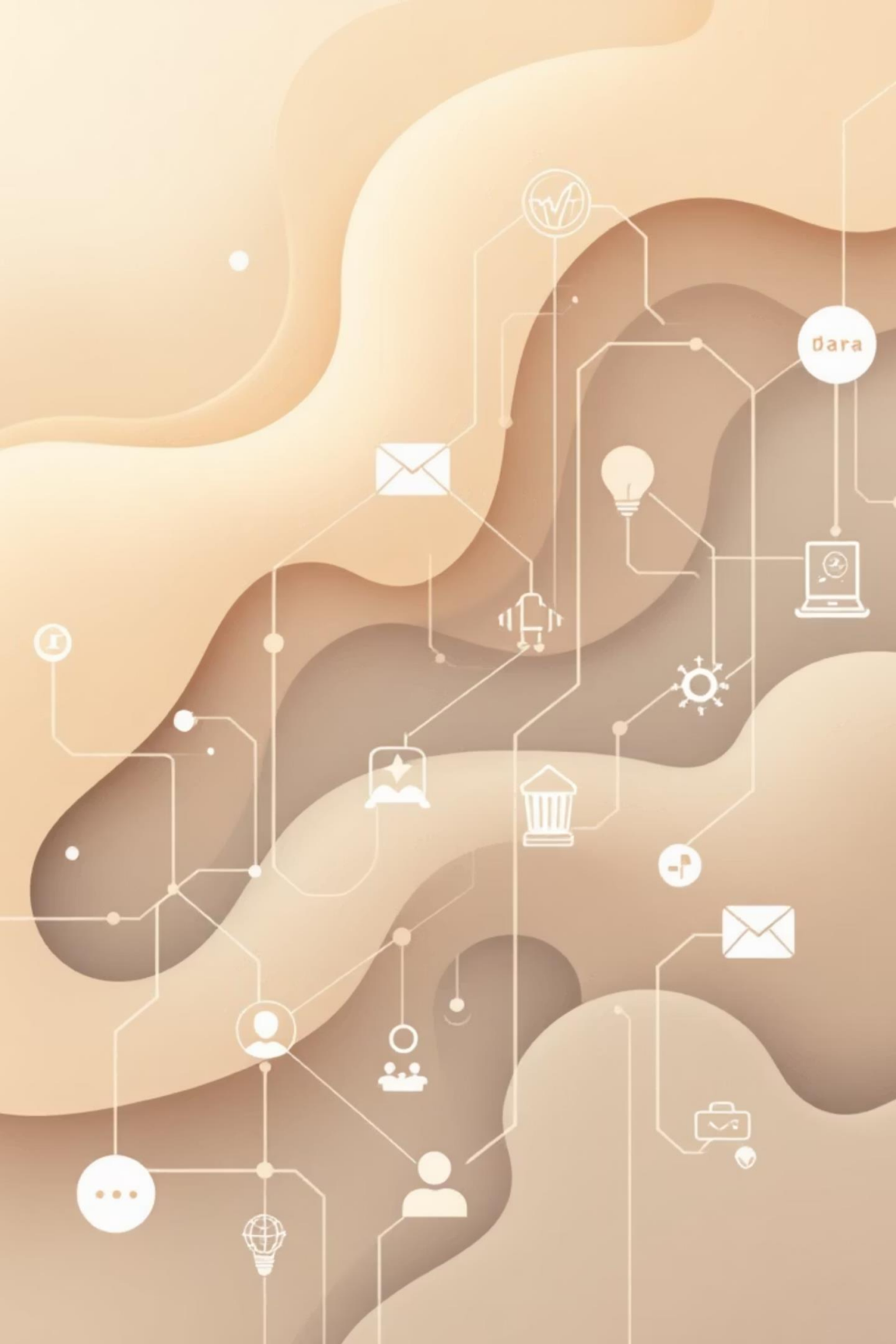
- **Boston (MBTA):** Avg Delay 1.30 min, significant snow delays.
- **Chicago (CTA):** Avg Delay 1.31 min (highest), most severe winter impact.
- **New York (MTA):** Avg Delay 1.30 min, similar snow impact.

All cities show similar weather resilience patterns, with Chicago experiencing the highest overall delays.

Seasonal Patterns

- **Winter:** 1.45 min avg delay, 23% more than summer.
- **Summer:** Most reliable service.

Snowfall is the strongest predictor of delays (correlation 0.45).



Technical Implementation



Data Processing

Python, Pandas, NumPy, SciPy for analysis.



Database

PostgreSQL 13 with SQLAlchemy.



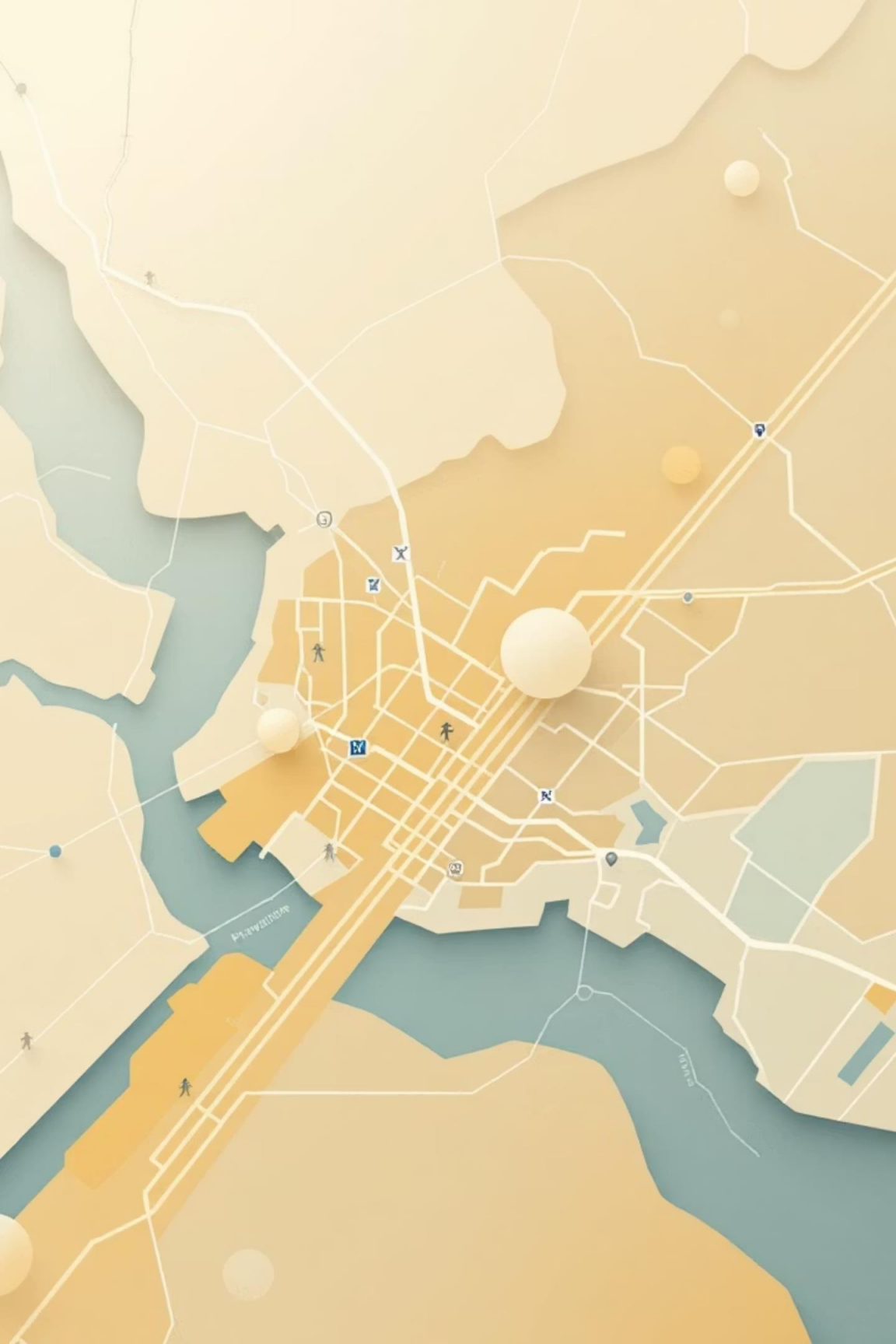
Orchestration

Apache Airflow 2.7.0, Docker for automated scheduling.



Visualization

Matplotlib, Seaborn, Jupyter, Power BI for dashboards.



Key Insights & Recommendations

Operational

Snow Events: Deploy 25-30% more vehicles, pre-position crews, add 15-20 min buffer.

Strategic

Infrastructure: Prioritize Chicago (heated shelters), Boston (drainage), study NYC resilience.

Communication

Proactive Alerts: Mobile app integration for weather-based delay predictions.



Business Value & ROI

\$700K

Annual Savings

Potential savings up to \$1.95M across 3 cities.

10

On-Time Performance

Points improvement in on-time performance.

20%

Satisfaction

Increase in passenger satisfaction.

This project transforms reactive crisis management into proactive strategic planning, with significant financial and service quality benefits.



Conclusion & Future Enhancements

Summary

- Quantified weather impact (snow = 114% more delays).
- Analyzed 364,038 records across 3 cities.
- Built production-ready automated pipeline.
- Delivered actionable insights and dashboards.

Future Enhancements

- **Short-Term:** Add more cities, real-time weather API, ML predictions.
- **Long-Term:** Predictive maintenance, dynamic route optimization, climate impact.